Problem Statement

Problem

People struggle to judge what is in their food, whether it is safe to eat, and what they should eat for their health goals. Nutrition labels are missing for home-cooked and restaurant food, spoilage is hard to judge by eye, restaurant hygiene is hard to verify, and generic diet plans ignore a person's age, goals, allergies and medical conditions.

Proposed solution

FoodVerse AI is one platform that takes images, video, audio and text and returns food insights across five modules.

Module	What it does
Food Recognition (Food Lens)	Identifies the dish from a photo, with optional voice or text context, and estimates serving size, calories, protein, carbs, fat and a health score
Freshness Detector	Judges whether food is fresh or spoiled
Hygiene Monitor	Checks images or video for hygiene and safety issues
Smart Kitchen	Gives cooking and recipe help
Meal Planner	Generates a personalized meal plan from the user's profile, goals, allergies and medical conditions using a local LLM

Target users

- Health-conscious individuals
- People with allergies or medical conditions who need safe meal plans
- Home cooks
- Restaurant owners and inspectors (hygiene)

Expected outcome

A working web app where a user uploads a photo or enters details and receives instant, explainable food insights.